Context-Aware Sinhala Post OCR Error Correction Model for Assistive Document Reading

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Project Proposal Report

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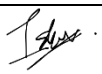
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The visually impaired people in Sri Lanka do not have a cost-effective and dependable way of accessing the printed Sinhala texts like newspapers and textbooks. Such content can be digitised using OCR technology, but even the most sophisticated Sinhalese OCR systems introduce a large number of character and word errors, and these errors are realised when the text is verbalised in a voice-based assistive system.

There is no currently existing transformer based context-aware post OCR correction system in Sinhala. The current methods are limited to rule based engines and dictionary look up procedures where each word is considered individually and they are trained on typographical errors made by humans and not on the visual noise patterns of OCR scanning of Sinhalese script.

This research proposes a two-stage correction model, which consists of a SinBERT error token detector and a fine-tuned mT5 sequence-to-sequence corrector, and the first domain-specific Sinhala OCR noise dataset, which classifies five types of errors inherent to the Sinhala script. It is expected that the error rate of character and word will be reduced, and the over-correction will be controlled, thus the OCR output of the text will be reliable enough to use in real-time assistive reading.

This has a direct social benefit to the Sinhala communities with low literacy and visually impaired, which contributes to the Sustainable Development Goals 4 and 10. The module can be deployed commercially as a standalone Sinhala document digitisation API to digitisation projects of government documents.

Keywords: Sinhala OCR, Post-OCR Correction, SinBERT, mT5, Assistive Technology, Low-Resource NLP, Error Correction

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Finally, I would like to thank my fellow group members for their contribution to the overall structure and development of the system. Although the group collectively decided the general scope and system components, each member independently worked on developing and improving their assigned parts. This approach helped ensure a fair distribution of work while allowing each member to focus on their specific area.

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LIST OF ABBREVIATIONS

OCR - Optical Character Recognition

NLP - Natural Language Processing

TTS - Text-to-Speech

STT - Speech-to-Text

CER - Character Error Rate

WER - Word Error Rate

RAG - Retrieval Augmented Generation

LLM - Large Language Model

MAT - Medial Axis Transform

WBS - Work Breakdown Structure

SDG - Sustainable Development Goal

BERT - Bidirectional Encoder Representations from Transformers

ZWJ - Zero Width Joiner

1.INTRODUCTION

1.1 Background and Context

As stated by the National Survey of Blindness and Visual Impairment in Sri Lanka (2014-2015) [1], about 150,000 individuals are blind and there were 400,000 with low vision which comprise 17% of the total population affected by some type of visual impairment. The situation of these communities is that access to Sinhalese newspapers, textbooks and government notices continues to be an obstacle to education, employment and participation in social life, therefore directly contradicting SDG 4 (Quality Education) and SDG 10 (Reduced Inequalities).

One possible solution is to use optical character recognition (OCR), which transforms the scanned documents images into text readable by the machine and then it can render the text as voice text. But Sinhalese is morphologically complicated and resource poor [2]. vowel modifiers are placed around consonants in several places, a 'hal-sign' represses vowel sounds, and characters of ZWJ compounds are frequent. These complications affect proper OCR recognition. Even the latest Sinhala OCR system achieves a word error rate of 19.57% on clean texts and this error rate is even worse when used on real newspaper scans with multi column formats and mixed font groups. Every character that is misrecognized in a voice based assistive system is read out to the user then output of uncorrected OCR is not only inaccurate but also misleading.

Current post-OCR correction systems of the Sinhalese language, which have been created between 2016 and 2021, are purely rule-based [4] and dictionary search methods. The tokens are processed individually, with no sentence level context, and the systems were optimized to reduce human typing errors as compared to the visual noisy patterns created by OCR. So far, no Sinhalese OCR rectification model has the learning based or contextual information. This system fills this gap by proposing a two stage transformer correction framework that combines SinBERT to detect mistakes and mT5 to fix them using sentence context as a component of an assistive reading pipeline running in real time. Finally the system will ensure that what a visually impaired user hears is accurate and reliable.

1.2 Literature Review

1.2.1 Sinhala-Specific Correction Systems

Previous research on postprocessing of Sinhalese OCR had been carried out earlier at the University of Colombo in 2016 [4], the first works to combine postprocessing with Tesseract OCR were done. This method was based on Unicode normalization and a collection of manually handwritten substitution rules. In 2018, a more formal approach became known as morphological dictionary looking mechanism developed at the University of Moratuwa , with prefix root suffix decomposition and minimum edit distance. According to this system, there was a significant increase in word accuracy in that it increased by 59.8% to 92.6% [5] on a corpus of controlled documents. The common factor in all the three systems is that each of them has a basic limitation they do the correction on a word-by-word basis without taking into account the context of the sentence. This means that the rule sets cannot possibly generalise to previously unobserved patterns and none of the methods has been tested on newspaper layouts or documents with more than one column.

The most commonly deployed Sinhalese correction tool is SinSpell (2021) [6], which is a Hunspell-based engine that is also available as a web service. An extension of this engine in 2024, using permutation based candidate generation and edit distance ranking, was based on data[6]. However, both versions address human typing errors instead of visual noise that OCR generates, the latter of which have a completely different error distribution. Both systems do not differentiate between phonetically similar characters, which is a relevant issue when it comes to spell checking, and visually similar characters,[7] which are important when it comes to OCR correction.[6]

1.2.2 Transformer-Based Sinhala NLP

Sinhala natural language processing methods based on transformers have evolved fast. In 2022 SinBERT, a monolingual model derived based on RoBERTa, was pre-trained on 15 million of Sinhalese texts [8] and shown to have good textual context on a variety of Sinhalese NLP tasks. Its two way attention fits it well to detecting errors in tokens but, so far, it has not been used to detect OCR noise. In 2025, SinBERT was used in a pipeline that corrected Sinhala speech dyslexia [10] using mT5 and a large language model in a multi-stage system with 70% on its test set. The architecture is similar to that of the present proposal but the system is designed to deal with specific errors of speech-driven dyslexia, phonological substitutions and letter reversals, but not with OCR visual noise, and its training data do not conform to that domain.

1.2.3 Post-OCR Correction for Low-Resource Languages

The sequence-to-sequence models with transfer learning and post-OCR error correction of low-resource languages have shown that the character level sequence-to-sequence model can get significant CER improvement even in the case of limited data [11]. An experiment of benchmarking in 2025 [12] validated that fine-tuned sequence-to-sequence transformers are always better at post-OCR correction than a decoder only generative model, which governs the choice of the model [11]. The RoundTripOCR methodology (2024) used noisy-clean pair generation to overcome the data scarcity of six Devanagari languages, but not Sinhala, despite the similarity of the data scarcity issues [13]. In a benchmarking study of Sinhalese OCR in 2025, it was reported that Surya-OCR had word-error-rate of 19.57% and a CER of 2.58% on clean documents and significantly lower on newspaper layouts, indicating the need to have a dedicated correction layer, not an optional one. [12]

1.3 Research Gap

According to the literature, the available Sinhala correction schemes are always rule-based and word-based, focusing on typographical mistakes that human writers are likely to make, but focusing on visual artefacts that are added by OCR mechanisms. Though current academic developments with Sinhalese natural language processing have been made, no transformer-based context-aware post-OCR correction model is published yet.

The nearest initiative, 2025 Sinhala dyslexia correction pipeline, takes a similar architecture (SinBERT + mT5) [10]. Therefore this time user-specific architecture is adjusted to a different error domain rather than OCR. The errors in OCR are due to visual confusion between printed glyphs, diacritic deletion and compound character misrecognition, which have structural properties that are inappropriate to the phonologically determined dyslexia error patterns discussed by the mentioned model. therefore, it would require the extensive retraining of pretrained weights on OCR-specific data to apply it to the OCR domain.

Another gap is that there is entirely no domain specific Sinhalese OCR noise dataset, that includes newspapers and textbooks. Although RoundTripOCR methodology [13] has been able to reduce the problem of data scarcity in low-resource scripts through noise-clean synthesis of pairs, Sinhala has not been integrated into noise-clean synthesis efforts. There is also no special Sinhala OCR error classification. the past research has chunked together all the cases of mismatched characters and, therefore, did not evaluate the visual structure of Sinhalese writing that dictates the rule of appearance of certain types of errors in practice.

Lastly, the literature is also lacking the description of the requirements related to a real-time assistive reading pipeline where over-correction can be more harmful than the absence of corrections altogether because the wrongly read words are pronounced to the visually impaired users.

System	Approach	Context-Aware	OCR-Specific	Sinhala
Weerasinghe 2016	Rule-based + Unicode normalization	✗	✗	✓
Premasiri 2018	Morphological dictionary lookup	✗	✗	✓
SinSpell 2021	HunSpell spell checker	✗	✗	✓
Kashid et al.	RoundTripOCR noise generation	✓	✗	✗
Dyslexia pipeline 2025	SinBERT + mT5	✓	✗	✓
This research	**SinBERT + mT5 (OCR-trained)**	✓	✓	✓

Table 1 - Research papers gap comparison

2.OBJECTIVE

2.1 Main Objective

To develop a two stage context-aware post-OCR error correction system in Sinhala printed text, one needs to combine an SinBERT-based error detection system with an mT5-based sequence correction system, which translates into a quantifiable improvement in the CER and WER of OCR outputs obtained using Sinhalese newspaper and textbook materials. The aim is to give confidence in textual input into a voicebased, real time, assistive reading device, which is expected to be used by the visually impaired.

2.2 Specific Objectives

2.2.1 Define a Sinhala-Specific OCR Error Taxonomy

The initial goal is to perform a systematic study of the results of Tesseract and Surya-OCR applied to printed Sinhalese newspaper and textbook samples, and the goal is to classify the errors obtained. Up to now, there is no established typology of the Sinhalese OCR errors. To this end, the outcome of this objective will be a classification consisting of five categories. The design of Objective 2 synthetic noise generation will be directly informed by the classification.

Error Type	Description	Example
Vowel modifier omission	Diacritics (ා, ි, ී, ු, ූ) dropped by OCR	රාජ → රජ
Hal-sign misdetection	Hal-sign (්) missed or misplaced	ශ්‍රී → ශරී
Visually similar substitution	Lookalike consonants confused	ක → ඛ, ව → බ
Compound glyph segmentation	ZWJ-based compounds split or merged incorrectly	ශ්‍ර split into ශ්
Font encoding noise	Legacy FM encoding misread as Unicode values	Garbled characters from unsupported fonts

Table 2- Sinhala OCR Error Taxonomy - Five Category Classification

2.2.2 Construct a Domain-Specific Sinhala OCR Noise Dataset

The second objective is to build the first domain specific Sinhala post-OCR correction dataset as noisy-clean sentence pairs. No such dataset currently exists for Sinhala newspaper or textbook text. Real OCR pairs will be collected by running Lankadeepa and Mawbima newspaper pages through Tesseract and Surya-OCR and aligning the output with ground truth

text at sentence level. This will be supplemented with synthetic pairs generated using a noise injection engine calibrated to the error frequencies observed in Objective 1. The final dataset will contain approximately 120,000 sentence pairs split across training, validation, and test sets.

Split	Size (target)	Source
Training	~100,000 pairs	70% synthetic + 30% real
Validation	~10,000 pairs	Mixed
Test	~10,000 pairs	Real OCR noise only

Table 3 - Sinhala OCR Noise Dataset - Train/Validation/Test Split

2.2.3 Fine-tune SinBERT as an OCR Error Token Detector

Fine-tune SinBERT-Large as a binary token-level classifier that identifies OCR-corrupted tokens within their sentence context. SinBERT is a RoBERTa-based monolingual Sinhala model pre-trained on 15 million Sinhala texts [8], making it well suited to judge whether a token is plausible given its surrounding context. A token classification head will be added and trained on the noise dataset from Objective 2, with class weighting applied to handle the natural imbalance between correct and error tokens. A confidence threshold will also be tuned to prevent over-flagging of valid but rare tokens such as proper nouns.

2.2.4 Fine-tune mT5 as a Context-Aware Sequence Corrector

Fine-tune mT5-small as a sequence-to-sequence corrector that takes sentences with flagged error positions as input and generates corrected Sinhala text as output. mT5 is a multilingual text-to-text transformer pre-trained across 101 languages [9]. Treating OCR correction as a translation task, noisy text to clean text. allows mT5 to generate contextually consistent corrections rather than fixing tokens in isolation. Critically, mT5 only processes sentences where SinBERT has flagged at least one token, which significantly reduces the risk of introducing new errors into already correct text.

2.2.5 Evaluate the Full Two-Stage Pipeline Against Baselines

Evaluate the full SinBERT + mT5 pipeline against three baselines on the real OCR test set. Results will be reported both as aggregate metrics and broken down per error category from the classification, to show clearly which error types the system handles well and where limitations remain. Inference latency on a standard laptop CPU will also be measured to confirm the pipeline is suitable for real-time use in the assistive reading context.

3. METHODOLOGY

3.1 Research Approach

The proposed research will adopt a Design Science Research (DSR) approach. This adds another layer of challenge to working with DSR frameworks, as it is common in applied computing research that the goal is not merely understanding a problem but also creating something which solves that problem. The two stage correction model and the Sinhala OCR noise dataset constructed to train it is the artifact in this case.

3.2 Data Collection and Dataset Construction

In fact, since there is no Sinhala post-OCR correction dataset in existence, creating one of those sets is not just a preparation for this research, it is a core part.

3.2.1 Primary Data - Real OCR Noise Pairs

The pages of Mawbima and Lankadeepa Sinhala newspapers are scanned and passed through Tesseract and 'Surya-OCR'. If the original digital text is available, the OCR output will be aligned with it at sentence level to create ground truth noisy-clean pairs. Such pairs will reflect true error patterns of OCR for real newspapers and their corresponding (font, layout) not something that can be recreated with synthetic data only.

3.2.2 Secondary Data - Synthetic Noise Generation

Clean Sinhala text scraped from verified news portals and digital textbooks from the Educational Publications Department (edupub.gov.lk) will be used to generate additional noisy-clean pairs using the RoundTripOCR approach adapted for Sinhala.[13] Noise will be injected programmatically based on the error taxonomy defined in Objective 1. covering diacritic drops, hal-sign removal, visually similar character substitution, compound character splits, and spacing errors. with injection probabilities calibrated against observed real OCR error frequencies to ensure the synthetic pairs reflect genuine scanner behaviour.

The final training set will combine 70% synthetic and 30% real pairs. The test set will contain real OCR noise only, ensuring that evaluation results reflect actual performance on genuine documents rather than artificially clean conditions.

3.2.3 Ethical Considerations

All newspaper content used for data collection is publicly available and no personal data is involved at this stage. Voice interaction data collected during user testing will be handled with informed consent, anonymised before storage, and managed in line with the data protection principles outlined in the project's ethical awareness framework.

3.3 Technical Methods and Algorithms

3.3.1 Error Token Detection (SinBERT)

SinBERT-Large is a binary token classifier that has been fine-tuned. The model predicts CORRECT or ERROR, which is based on the suitability of a token to its environment, reading the entire sentence before every prediction, through bidirectional attention. In training, cross-entropy loss is applied with a weight on the classes to offset the imbalance, the majority of tokens are correct and this would bias prediction towards CORRECT. Only tokens with the flag ERROR proceed to Stage 2. This design eliminates over-correction, as the model that is used to correct does not touch tokens that are seen as clean by the detector.

3.3.2 Context-Aware Correction (mT5)

mT5-small is optimized as a sequence-to-sequence correcter. The model is fed with the full sentence with the errors highlighted and then produces the fixed version of the text and this is because first, it has been proven previously that mT5 -small can perform well on Sinhalese text generation; second, because its subword tokenizer can handle Sinhala Unicode intuitively without splitting characters; lastly, because the small variant can run well on the local hardware available, which is extremely vital since this model is required to operate without relying on the cloud. ByT5 was also taken into consideration since its tokenization at the byte level is also good with diacritics but the multilingual pre-training of mT5-small provides a better base knowledge in the Sinhala language and the tokenization advantage in this case is surpassed by the pre-training.

3.3.3 Preprocessing

All Sinhala text will be normalized to Unicode NFC form before being passed to any model. This is a compulsory step. text scraped off different web pages may have sequences of identical appearance, but different in Unicode, which corrupts training without remedy unless detected. Any FM-encoded text encountered in the collection of data is converted to Unicode prior to being placed in the pipeline.

3.4 Technology Stack and Justification

Tool / Library	Purpose	Justification
HuggingFace Transformers	Model fine-tuning and inference	Industry standard for transformer fine-tuning. native support for SinBERT and mT5
SinBERT-Large	Error token detection	Best available Sinhala contextual language model. pre-trained on 15M Sinhala texts
mT5-small	Sequence-to-sequence correction	Confirmed effective for Sinhala generation. computationally feasible. subword tokenizer handles Sinhala Unicode
Tesseract / Surya-OCR	Generating real OCR noise pairs	Best available Sinhala OCR engines [3][12] for dataset construction
Python / PyTorch	Model training pipeline	Standard deep learning framework. full HuggingFace compatibility
Google Colab	Training compute	Free GPU access sufficient for mT5-small fine-tuning
BeautifulSoup	News corpus scraping	Reliable tools for structured web scraping from Sinhala news portals
Unicodedata	NFC normalization	Standard library for Unicode normalization. critical for Sinhala text consistency

Table 4 - Technology Stack and Justification

3.5 Validation Strategy and Evaluation Metrics

The model is evaluated against four baselines across three primary metrics

3.5.1 Baselines

System	Description
Baseline 1	Raw OCR output - no correction
Baseline 2	Rule-based correction replicating
Baseline 3	Single-stage mT5 without SinBERT detector
Proposed	Two-stage SinBERT + mT5 full pipeline

Table 5 - Evaluation Baselines

3.5.2 Metrics

Metric	Definition
CER	Character Error Rate. proportion of characters wrong
WER	Word Error Rate. proportion of words wrong
Over-correction Rate	Proportion of correct tokens incorrectly changed
Inference Latency	Processing time per sentence on laptop CPU

Table 6 - Evaluation Metrics

Results will be reported per error category to show which error types the model handles best and where limitations remain. a level of analysis that goes beyond simple aggregate metrics and demonstrates genuine understanding of the problem.

4.HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overall System Overview

The full system is a glasses based assistive document reader comprising four integrated components working in sequence. A camera mounted on a pair of glasses captures images of physical Sinhala documents, which are processed through the pipeline and delivered as natural Sinhala voice output to the user.

The four components and their roles within the pipeline are,

Component	Role
Component 1 - Topological Sinhala Character Reader	Layout detection and OCR of scanned document
Component 2 - Post-OCR Error Correction (This work)	Cleans and corrects OCR output before retrieval
Component 3 - Advanced Document Retrieval and RAG	Retrieves relevant content and generates responses
Component 4 - Adaptive Voice Interaction Module	Handles voice input/output and user personalization

Table 7 - Full System Component Roles

Component 2 acts as the quality gate of the pipeline. when its output is erroneous, then all the subsequent components are negatively impacted. Errors that remained after correction are read out to the visually impaired user making correction accuracy an important system requirement.

4.2 Full System Architecture Diagram

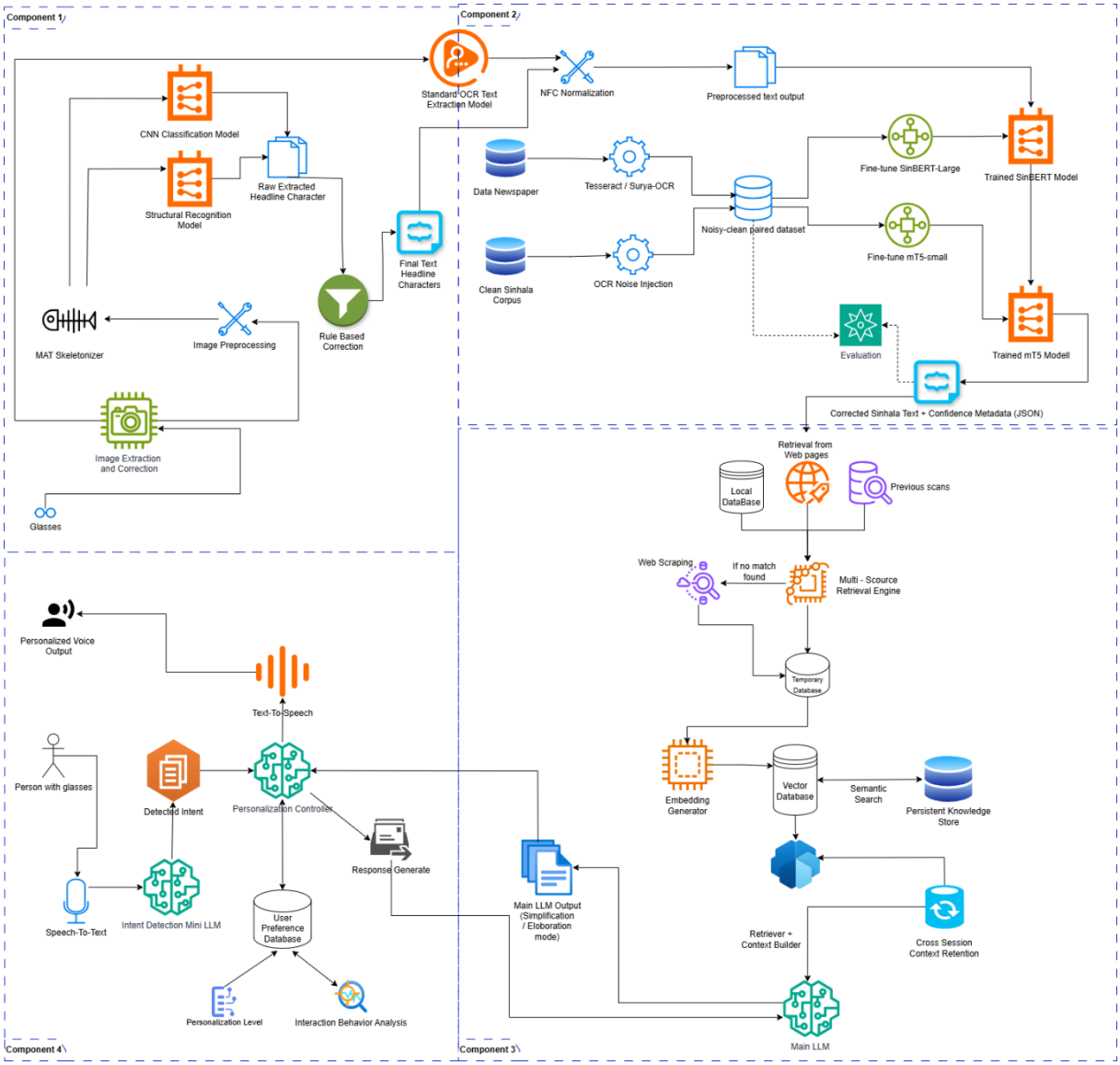


Figure 1- Full System Architecture Diagram

4.3 Component 2 - Detailed Architecture

This section describes the internal architecture of the post-OCR error correction component in detail, covering its two stages, data flow, and integration interfaces.

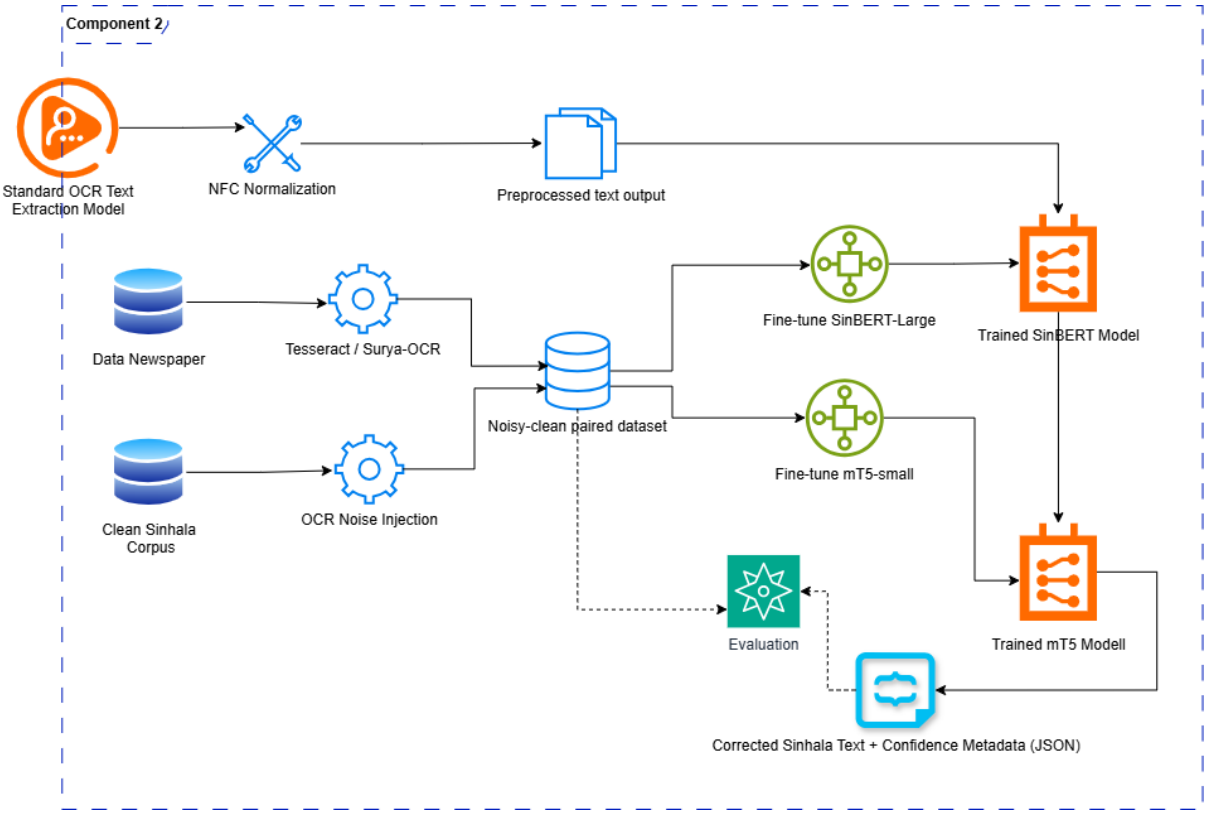


Figure 2 - Context-Aware Sinhala Post OCR Error Correction Model Detailed Architecture

4.3.1 Internal Architecture

The component operates as a two-stage pipeline. Stage 1 detects which tokens in the OCR output are likely corrupted. Stage 2 corrects only those flagged tokens using their surrounding sentence context. This separation is deliberate, it prevents the correction model from altering tokens that are already correct, which is the primary cause of over-correction in single-stage systems.

4.3.2 Data Flow

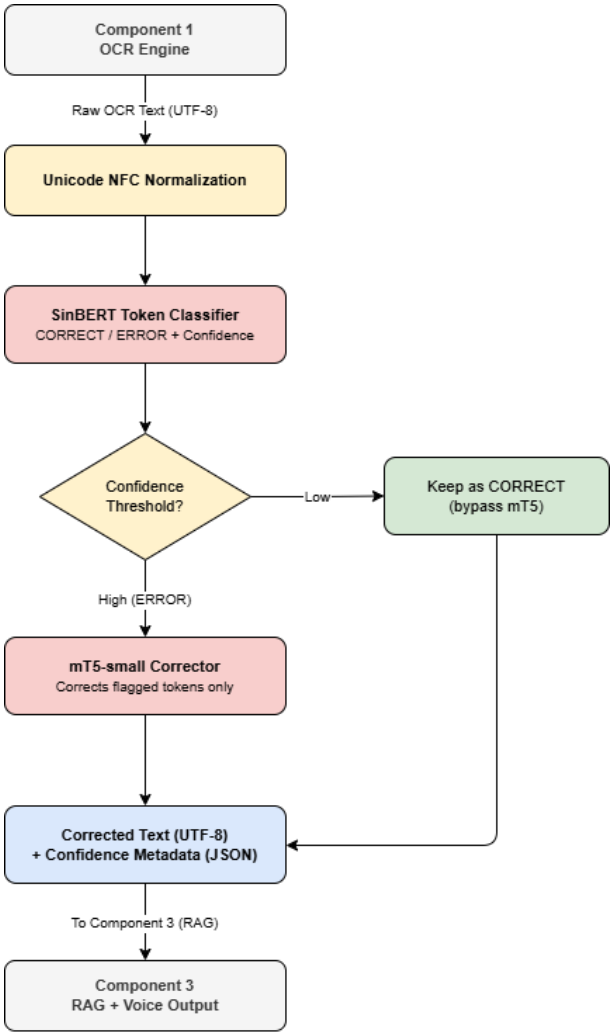


Figure 3 - Data Flow Diagram

1. **Input received** from Component 1 - raw Sinhala text string extracted from the scanned document, and may contain errors added by optical character recognition.
2. **Unicode normalization** applied - all text converted to NFC form before any model processing
3. **SinBERT detection** - full sentence passed to the fine-tuned SinBERT token classifier. each token receives a binary label and a confidence score
4. **Threshold filtering** - Tokens with a low confidence are held back as CORRECT, thus avoiding the over-flagging of ambiguous or rarely used tokens like proper nouns
5. **mT5 correction** - entences that have at least one flagged token are sent to mT5 where the incorrect positions are explicitly labeled. the model generates a fixed sentence
6. **Output delivery** - corrected text string passed to Component 3 along with per-token confidence metadata

4.3.3 Integration Interfaces

The OCR engine produces raw Sinhala text which is sent between Component 1 and Component 2 as a UTF 8 string so that it can be used to detect and fix errors. Component 2 sends the fixed text to Component 3, once again in the form of a UTF-8 string, and it is ready to be further processed by voice rendering or the next processing. Besides the fixed text, Component 2 also sends per-token confidence scores in a form of a JSON object, which Component 3 can use to handle corrections of uncertain output accordingly instead of assuming a uniform confidence level in all output.

4.4 Scalability and Security Considerations

Scalability - The module can process one paragraph at a time, which is adequate to the current case of newspaper and textbook. To extend to larger document types or greater throughput, the model can be exported to ONNX format to execute it faster at a faster and more efficient CPU without retraining or batched processing can be applied at the document level.

Security and Privacy - No images, audio or personal data are passed through the correction module; all the data is text only. Data in voice interaction received at Component 4 is not transmitted to this module. The entire processing is done on the user laptop without any external API calls and thus the user data privacy and offline functionality are guaranteed.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The main stakeholders are the visually impaired people in Sri Lanka with the first language of Sinhalese. This is the most pressing concern in the system as the soundness of the output that comes out of the system is fully based on the effectiveness of the correction module. This means that they have their needs given priority in every design.

The second group of the stakeholders includes older and low-literacy Sinhalese speaking who, although may not be visually impaired, have issues with printed materials. The voice output of the assistive reading system performs a similar role to these users, and proper correction is equally important to their overall experience.

Government agencies involved in massive document digitisation, including the National Archives and the Ministry of Education, will gain access to a stable, automated correction layer, which will improve Sinhalese OCR performance without requiring any human intervention. In this regard, Sinhala media organisations and digital publishers show the same interest, especially in relation to enhance the quality of the digitised content archives.

Lastly, Sinhalese NLP research community is another group of stakeholders. The OCR noises dataset and its taxonomy of errors produced by this study will be the first resources of their kind in Sinhalese; it will be a reusable resource on which researchers in future empirical research on low-resource language processing and assistive technologies can base their work.

5.2 Functional Requirements

1. The system will take raw Sinhalese OCR text as input and produce correct Sinhalese text as output.

2. The system will detect OCR-corrupted tokens in a sentence using a SinBERT-based binary token classifier, which is fine-tuned on Sinhala OCR noise data.

3. The system will only fix the tokens that are marked as erroneous with the help of an mT5-based sequence-to-sequence model, and all the rest of the tokens will remain intact.

4. The system will be able to handle all five categories of predefined Sinhala OCR error: vowel modifier omission, hal-kiri misdetection, visual similar substitution, compound glyph segmentation, and font-encoding noise.

5. Before processing, the system will normalize all input text to Unicode Normalization Form (NFC) to provide stable tokenisation.

6. The system will give a per-token confidence score with the fixed-text so that it can serve downstream pipeline components.

7. The system will be connected to Component 1 (OCR), and Component 3 (RAG) through specified UTF-8 text interfaces.

5.3 Non-Functional Requirements

Performance - The correction module will be able to process a typical newspaper paragraph (100 - 150 tokens) in three seconds on an average laptop CPU, which meets real-time constraints of assistive reading pipeline.

Accuracy - The system must show statistically significant improvement in both CER and WER compared to uncorrected OCR output as tested on the held out real OCR noise test set.

Over-correction Control - The percentage of correctly spelled tokens which are erroneously changed (overcorrection rate) will be lower than the percentage of the single stage mT5 baseline, as determined by the established evaluation metrics.

Privacy - All text processing will be done on the device, and no external API calls will be made, which will guarantee that user data will never leave the device.

Reliability - The system will be resistant to bad input (e.g. empty strings, bad characters) by not crashing, and returning the original input when it is not possible to do a reliable correction.

Maintainability - Model weights and configurations should be stored separately of the pipeline code, so that it is possible to update the model without having to change integration interfaces.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The main focus market includes institutions and individuals in Sri Lanka that are involved with the Sinhalese printed documents in large quantities, i.e. government departments, media publishers, universities, and providers of accessibility services. A secondary market involves visually impaired end users and their families who need to have access to Sinhalese document-reading tools at affordable prices. A persona of an early adopter is associated with a medium sized government department or a Sinhala newspaper that is currently using manual proofreaders to fix OCR output and wants to save money by automating the process.

6.2 Value Proposition

The Sinhalese OCR correction tools that are currently in use are either rule based with low accuracy or costly enterprise licenses that do not support the Sinhalese language. The current component presents the first learning based and context-aware Sinhala OCR correction model, accurate, cost-effective, locally engineered, and offline. In the case of visually impaired users, the value proposition is obvious. the audio output is a reflection of the original text as opposed to a sequence of misread characters that may distort the meaning. In the case of institutional clients, the benefit is a plug-and-play correction API, which improves the quality of any Sinhalese OCR pipeline without requiring changes in existing infrastructure.

6.3 Revenue Model

Assistive Device Subscription

The assistive reader is a four component glasses-based system (full system) that is sold as a low price hardware package with a monthly software subscription. The correction module is one of the main parts of this product. Target price: LKR 2500-3500/month - much lower than imported options like OrCam MyEye, which retails at USD 2000 at once.

OCR Correction API Licensing

The correction model is an API service under a REST license to government departments, publishing houses and media organisations to use in document digitisation processes. The price is determined by volume rates.

Dataset Licensing

The Sinhalese OCR noise data set that was created as part of this study is a one of its kind. It may be licensed to academic institutions, NLP research teams, and technology firms that develop Sinhala language technology a model that has already been effectively used by datasets like Common Voice and RoundTripOCR derivatives.

7. BUDGET AND JUSTIFICATION

This research does not need any special hardware acquisition. The implementation and fine-tuning of the models will be done on an existing personal laptop together with Google Colab Pro to run the tasks that require a great deal of the GPU, therefore eliminating the need to have a separate workstation. All the central software, such as Python, PyTorch, HuggingFace Transformers, Tesseract OCR, and BeautifulSoup, are open-source, and thus there are no license fees.

The main recurrent cost is the subscription to Google Colab Pro that is required to train SinBERT and mT5-small outside the memory capacity of the free tier. Other distributions go to cloud storage to version datasets, internet connection to collect data.

Category	Item	Estimated Cost	Justification
Hardware (External)	Camera, Microphone	LKR 7,000 upwards	These are essential to build the system and even if we use pre-shot images, the better camera directly improves the project results.
Software	All tools open-source (Python, OpenCV, scikit-image) Google colab pro	LKR 9,000	All primary tools are freely available
Data	Print media from website, capture images	Depends, as documents to capture images can be bought or borrowed.	Ground truth dataset creation
Man-Hour	Student research time	-Not applicable	Already accounted for in the project
Hardware (Internal)	Computer	-Not applicable	Current available laptops will most be able to the software.

Table 8 - Budget Breakdown

8. WORK BREAKDOWN STRUCTURE (WBS)

WBS	Task	Description	Deliverable	Duration
1	Literature Review	Review Sinhala OCR, SinBERT, mT5 and low-resource NLP papers	Literature summary + research gap table	5 weeks
2	OCR Error Taxonomy	Analyse OCR errors on newspaper pages, classify into 5 types	Taxonomy document	2 weeks
3.1	Real OCR Dataset	Scan newspapers, align OCR output with ground truth	~30,000 real noisy-clean pairs	3 weeks
3.2	Synthetic Dataset	Inject noise into clean Sinhala text using RoundTripOCR	~90,000 synthetic pairs	3 weeks
3.3	Dataset Finalisation	Merge, normalize and split into Train/Val/Test	Final dataset	1 weeks
4	SinBERT Fine-tuning	Train token classifier to detect OCR error tokens	Trained SinBERT model	4 weeks
5	mT5 Fine-tuning	Train seq2seq model to correct flagged tokens	Trained mT5 model	5 weeks
6	Pipeline Integration	Connect all stages, build Component 1 and 3 interfaces	Working pipeline + interface code	4 weeks
7	Evaluation	Measure CER, WER and over-correction against 3 baselines	Final evaluation report	4 weeks
8	Report Writing	Write progress reports, final report and research paper	All submission documents	3 weeks

Table 9- Work BreakDown Structure

9. GANTT CHART

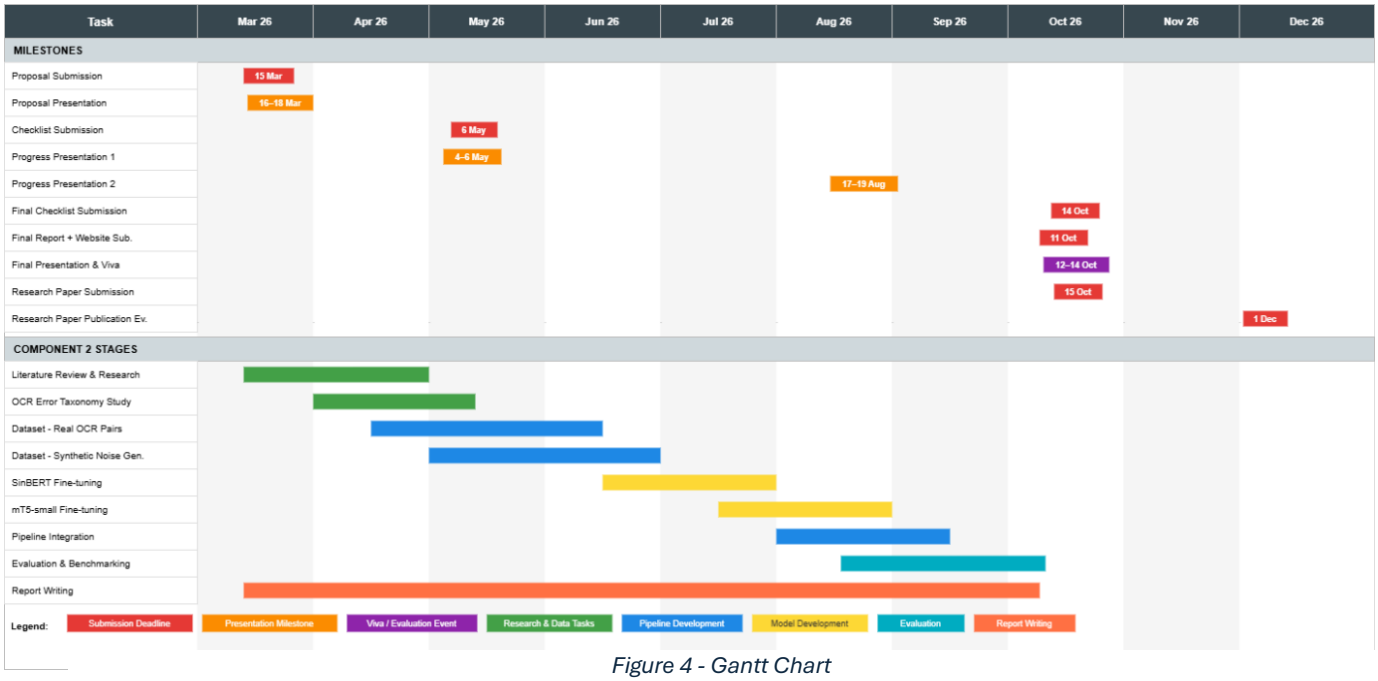


Figure 4 - Gantt Chart

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